

ILYES JAMOUSSI

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EDUCATION

Polytechnique Montréal

Montreal, CA

Bachelor of Engineering in Software Engineering – Artificial Intelligence & Data Science Concentration

Jan. 2023 – Expected Dec. 2027

- Relevant Coursework: Machine Learning, Reinforcement Learning, Advanced Algorithms, Statistical Learning

Pierre Mendès France High School

Tunis, TUN

French Baccalaureate – Mathematics & Computer Science, “Mention Très Bien” (highest honors)

Sep. 2018 – Jun. 2022

EXPERIENCE

Calcul Québec – Quantum Computing Summer Program

Montreal, CA

Researcher – Hybrid Quantum Algorithms

Jun. 2026 – Present

- Selected for a mentored research program with hands-on access to MonarQ, a 24-qubit universal quantum computer, and GPU/CPU high-performance computing clusters.
- Developing hybrid quantum-classical algorithms in Python, integrating quantum circuit execution with classical optimization routines.

AWS – Amazon Web Services

Montreal, CA

Cloud Consultant Intern – ProServe A2C Team

May. 2025 – Aug. 2025

- Developed an iOS & Android app that replaced paper workflows for 100+ technicians, cutting reporting delays from months to seconds.
- Architected a serverless backend leveraging AWS Cloud services, auto-scaling to 1M+ users and ensuring global availability.
- Implemented offline-first functionality with AWS AppSync, allowing offline use with automatic data synchronization.
- Integrated AI assistant using Amazon Bedrock Claude Sonnet 3.5, streamlining field documentation across technicians.

Polytechnique Montréal

Montreal, CA

Teaching Assistant – Data Structures and Algorithms

Jan. 2025 – Apr. 2025

- Guided students in implementing data structures in Java, including hash maps and AVL trees, while ensuring optimal Big O complexity.
- Taught algorithms in sorting, graph theory, and dynamic programming, such as Bellman-Ford and Rabin-Karp.

Polytechnique Montréal – Student Services (SEP)

Montreal, CA

Team Lead – Peer Tutoring

May 2025 – Present

- Authored the program’s first official Tutor Guide, adopted for all tutors starting Fall 2026, codifying academic-integrity standards.
- Led tutor recruitment, interviewing candidates and defining the criteria used to evaluate teaching ability.

Peer Tutor

Oct. 2024 – Apr. 2025

- Tutored students in Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Discrete Mathematics, and Design Patterns.

EXTRACURRICULARS

Accenture AOTC Program – AI Strategy Mandate

Montreal, CA

Consultant | CCGP – Polytechnique Management Consulting Committee

Jan. 2025 – Apr. 2025

- Designed RAG-powered chatbot for 10,000+ document portal, reducing navigation time by 75% for 1,000+ users.
- Recommended AI workflow automation and implementation roadmap, reducing 2+ hours daily per employee on repetitive tasks.

PolyAI – Polytechnique’s AI Society

Montreal, CA

Event Manager

Jan. 2024 – Dec. 2024

- Orchestrated AI events including hackathons, learnathons, and expert panels, engaging 200+ students per event.
- Secured partnerships with 8 tech sponsors, generating \$15,000+ in funding within 6 months through strategic outreach.

PROJECTS

MiniGPT – Generative Language Model | [Code](#)

- Trained a ~14M-parameter decoder-only GPT in PyTorch on TinyStories with BPE, reaching 5.32 validation perplexity.
- Served token-streaming inference via FastAPI and a live Streamlit demo; Dockerized with CI checks using pytest, ruff, and mypy.

MailGuard AI – Email Spam Detector | [Code](#)

- Built an encoder-only Transformer in PyTorch with padding-masked attention and a custom 30K-token vocabulary.
- Achieved 97% accuracy and 0.95 spam F1 on SpamAssassin using stratified splits and class-weighted loss.

Online Multiplayer Strategy Game | [Code](#)

- Developed a real-time RPG multiplayer game with Angular frontend and NestJS backend leveraging REST APIs and WebSockets.
- Implemented GitLab CI/CD pipelines with automated builds, linting, 100% unit test coverage, and deployment on AWS EC2.

AVR Microcontroller Robot – Embedded Systems | [Code](#)

- Engineered robot in C++ using Dijkstra’s algorithm to follow black lines for shortest path navigation, stopping 10cm from target posts.
- Built with AVR ATmega324PA microcontroller, integrating infrared sensors, PWM motor control, I2C EEPROM, and UART communication.

CERTIFICATIONS

AWS Certified Developer – Associate | [Credential](#)

Aug. 2025

AWS Certified Solutions Architect – Associate | [Credential](#)

Jul. 2025

AWS Certified AI Practitioner | [Credential](#)

Jun. 2025

SKILLS

- ML & Data: Python, PyTorch, Transformers, scikit-learn, Pandas, NumPy, SQL
- MLOps & Engineering: Docker, FastAPI, GitHub Actions CI/CD, pytest, AWS (Bedrock, EC2), C++, Java, TypeScript, React, Node.js
- Spoken: English (Fluent), French (Native), Arabic (Native)